

The Art of Psychopharmacology

Potential Advantages

- Patients who have had past weight loss on either active compound alone may experience a more robust weight loss effect and better tolerability on the combination of naltrexone with bupropion than with administration of either drug alone

Potential Disadvantages

- Women who are pregnant or wish to become pregnant

Primary Target Symptoms

- Excess weight

Pearls

- No head-to-head studies with other drugs approved for weight management, so comparative efficacy to these agents is unknown; however, some patients who do not respond to these other agents may respond to Contrave
- Unlike phentermine/topiramate, naltrexone/bupropion does not have abuse potential
- Could theoretically be effective in binge eating disorder
- May be useful in antipsychotic-induced weight gain, especially for nonpsychotic patients taking antipsychotic agents for depression

- Contrave could theoretically exacerbate psychosis in patients with a psychotic disorder; however, no controlled studies of this use yet

Suggested Reading

Caixas A , Albert L , Capel I , Rigla M. Naltrexone sustained-release/bupropion sustained-release for the management of obesity: review of the data to date . *Drug Des Devel Ther* 2014 ;**8** :1419 –27.

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Wang GJ , Tomasi D , Volkow ND , et al. Effect of combined naltrexone and bupropion therapy on the brain's reactivity to food cues . *Int J Obes (Lond)* 2014 ;**38** (5):682–8 .

Nefazodone

Therapeutics

Brands

- Dutonin
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin receptor antagonist
- SARI (serotonin 2 antagonist/reuptake inhibitor); antidepressant

Commonly Prescribed for

(bold for FDA approved)

- **Depression**
- **Relapse prevention in major depressive disorder**
- Panic disorder
- Posttraumatic stress disorder (PTSD)

How the Drug Works

- Blocks serotonin 2A receptors potently
- Blocks serotonin reuptake pump (serotonin transporter) and norepinephrine reuptake pump (norepinephrine transporter) less potently

How Long Until It Works

- Can improve insomnia and anxiety early after initiating dosing
- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression

- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy, especially cognitive-behavioral psychotherapies, which have been specifically shown to enhance nefazodone's antidepressant actions
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer

Best Augmenting Combos for Partial Response or Treatment Resistance

* Venlafaxine and escitalopram may be the best tolerated when switching or augmenting with a serotonin reuptake inhibitor, as neither is a potent CYP450 2D6 inhibitor (use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation)

- Modafinil, especially for fatigue, sleepiness, and lack of concentration
- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, or treatment-resistant depression
- Benzodiazepines for anxiety, but give alprazolam cautiously with nefazodone as alprazolam levels can be much higher in the presence of nefazodone
- Classically, lithium, buspirone, or thyroid hormone

Tests

* Liver function testing is not required but is often prudent given the small but finite risk of serious hepatotoxicity

* However, to date no clinical strategy, including routine liver function tests, has been identified to reduce the risk of irreversible liver failure

Side Effects

How Drug Causes Side Effects

- Blockade of alpha adrenergic 1 receptors may explain dizziness, sedation, and hypotension
- A metabolite of nefazodone, mCPP (meta-chloro-phenyl-piperazine), can cause side effects if its levels rise significantly
- * If CYP450 2D6 is absent (7% of Caucasians lack CYP450 2D6) or inhibited (concomitant treatment with CYP450 2D6 inhibitors such as fluoxetine or paroxetine), increased levels of mCPP can form, leading to stimulation of 5HT_{2C} receptors and causing dizziness, insomnia, and agitation
- Most side effects are immediate but often go away with time

Notable Side Effects

- Nausea, dry mouth, constipation, dyspepsia, increased appetite
- Headache, dizziness, vision changes, sedation, insomnia, agitation, confusion, memory impairment
- Ataxia, paresthesia, asthenia
- Cough increased
- Rare postural hypotension



Life-Threatening or Dangerous Side Effects

- Rare seizures
- Rare induction of mania

- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)
- Rare priapism (no causal relationship established)
- Hepatic failure requiring liver transplant and/or fatal

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take once daily at night to reduce daytime sedation
- Lower the dose and try titrating again more slowly as tolerated
- Switch to another agent

Best Augmenting Agents for Side Effects

- Often best to try another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Many side effects cannot be improved with an augmenting agent
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of nefazodone

Dosing and Use

Usual Dosage Range

- 300–600 mg/day

Dosage Forms

- Tablet 50 mg, 100 mg scored, 150 mg scored, 200 mg, 250 mg

How to Dose

- Initial dose 100 mg twice a day; increase by 100–200 mg/day each week until desired efficacy is reached; maximum dose 300 mg twice

a day



Dosing Tips

- Take care switching from or adding to SSRIs (especially fluoxetine or paroxetine) because of side effects due to the drug interaction
- Do not underdose the elderly
- Normally twice daily dosing, especially when initiating treatment
- Patients may tolerate all dosing once daily at night once titrated
- Often much more effective at 400–600 mg/day than at lower doses if tolerated
- Slow titration can enhance tolerability when initiating dosing

Overdose

- Rarely lethal; sedation, nausea, vomiting, low blood pressure

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper is prudent to avoid withdrawal effects, but problems in withdrawal not common

Pharmacokinetics

- Half-life of parent compound is 2–4 hours
- Half-life of active mebatolites up to 12 hours
- Inhibits CYP450 3A4



Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant
- May interact with SSRIs such as paroxetine, fluoxetine, and others that inhibit CYP450 2D6
- * Since a metabolite of nefazodone, mCPP, is a substrate of CYP450 2D6, combination of 2D6 inhibitors with nefazodone will raise mCPP levels, leading to stimulation of 5HT_{2C} receptors and causing dizziness and agitation
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing nefazodone
- Via CYP450 3A4 inhibition, nefazodone may increase the half-life of alprazolam and triazolam, so their dosing may need to be reduced

by half or more

- Via CYP450 3A4, nefazodone may increase plasma concentrations of buspirone, so buspirone dose may need to be reduced
- Via CYP450 3A4 inhibition, nefazodone could theoretically increase concentrations of certain cholesterol lowering HMG CoA reductase inhibitors, especially simvastatin, atorvastatin, and lovastatin, but not pravastatin or fluvastatin, which would increase the risk of rhabdomyolysis; thus, coadministration of nefazodone with certain HMG CoA reductase inhibitors should proceed with caution
- Via CYP450 3A4 inhibition, nefazodone could theoretically increase the concentrations of pimozide, and cause QTc prolongation and dangerous cardiac arrhythmias
- Nefazodone may reduce clearance of haloperidol, so haloperidol dose may need to be reduced
- It is recommended to discontinue nefazodone prior to elective surgery because of the potential for interaction with general anesthetics

Other Warnings/Precautions

* Hepatotoxicity, sometimes requiring liver transplant and/or fatal, has occurred with nefazodone use. Risk may be one in every 250,000 to 300,000 patient years. Patients should be advised to report symptoms such as jaundice, dark urine, loss of appetite, nausea, and abdominal pain to prescriber immediately. If patient develops signs of

hepatocellular injury, such as increased serum AST or serum ALPT levels >3 times the upper limit of normal, nefazodone treatment should be discontinued

* No risk factor yet predicts who will develop irreversible liver failure with nefazodone and no clinical strategy, including routine monitoring of liver function tests, is known to reduce the risk of liver failure

- Use with caution in patients with history of seizures
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is taking an MAOI

- If patient has acute hepatic impairment or elevated baseline serum transaminases
- If patient was previously withdrawn from nefazodone treatment due to hepatic injury
- If patient is taking pimozide, as nefazodone could raise pimozide levels and increase QTc interval, perhaps causing dangerous arrhythmia
- If patient is taking carbamazepine, as this agent can dramatically reduce nefazodone levels and thus interfere with its antidepressant actions
- If there is a proven allergy to nefazodone

Special Populations

Renal Impairment

- No dose adjustment necessary

Hepatic Impairment

- Contraindicated in patients with known hepatic impairment

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension

Elderly

- Recommended to initiate treatment at half the usual adult dose, but to follow the same titration schedule as with younger patients, including same ultimate dose
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Safety and efficacy have not been established
- Preliminary research indicates efficacy and tolerability of nefazodone in children and adolescents with depression



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in

prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child
- For many patients this may mean continuing treatment during pregnancy
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Unknown if nefazodone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Trace amounts may be present in nursing children whose mothers are on nefazodone
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued

- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Depressed patients with anxiety or insomnia who do not respond to other antidepressants
- Patients with SSRI-induced sexual dysfunction

Potential Disadvantages

- Patients who have difficulty with a long titration period or twice daily dosing
- Patients with hepatic impairment

Primary Target Symptoms

- Depressed mood

- Sleep disturbance
- Anxiety

Pearls

- Preliminary data for efficacy in panic disorder and PTSD
- Fluoxetine and paroxetine may not be tolerated when switching or augmenting
- For elderly patients with early dementia and agitated depression, consider nefazodone in the morning and additional trazodone at night
- Anecdotal reports suggest that nefazodone may be effective in treating PMDD
- * Studies suggest that cognitive behavioral psychotherapy enhances the efficacy of nefazodone in chronic depression
- * Risk of hepatotoxicity makes this agent a second-line choice and has led to its withdrawal from some markets, including the withdrawal of Serzone from the US market
- Rarely, patients may complain of visual “trails” or after-images on nefazodone

Suggested Reading

DeVane CL , Grothe DR , Smith SL . Pharmacology of antidepressants: focus on nefazodone . *J Clin Psychiatry* 2002 ;**63** (1):10 –17 .

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Dunner DL , Laird LK , Zajecka J , et al. Six-year perspectives on the safety and tolerability of nefazodone . *J Clin Psychiatry* 2002 ;**63** (1):32 –41 .

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Khouzam HR . The antidepressant nefazodone. A review of its pharmacology, clinical efficacy, adverse effects, dosage, and administration . *J Psychosocial Nursing Ment Health Serv* 2000 ;**38** :20 –5.

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Masand PS , Gupta S . Long-term side effects of newer-generation antidepressants: SSRIs, venlafaxine, nefazodone, bupropion, and mirtazapine . *Ann Clin Psychiatry* 2002 ;**14** :175–82 .

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Schatzberg AF , Prather MR , Keller MB , et al. Clinical use of nefazodone in major depression: a 6-year perspective . *J Clin Psychiatry* 2002 ;**63** (1):18 –31 .

Nortriptyline

Therapeutics

Brands

- Pamelor
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin, norepinephrine reuptake inhibitor (SN-RI)
- Tricyclic antidepressant (TCA)
- Predominantly a norepinephrine/noradrenaline reuptake inhibitor

Commonly Prescribed for

(bold for FDA approved)

- **Major depressive disorder**
- Anxiety
- Insomnia
- Neuropathic pain/chronic pain

- Treatment-resistant depression

How the Drug Works

- Boosts neurotransmitter norepinephrine/noradrenaline
- Blocks norepinephrine reuptake pump (norepinephrine transporter), presumably increasing noradrenergic neurotransmission
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, nortriptyline can increase dopamine neurotransmission in this part of the brain
- A more potent inhibitor of norepinephrine reuptake pump than serotonin reuptake pump (serotonin transporter)
- At high doses may also boost neurotransmitter serotonin and presumably increase serotonergic neurotransmission

How Long Until It Works

- May have immediate effects in treating insomnia or anxiety
- Onset of therapeutic actions usually not immediate, but often delayed 2 to 4 weeks
- If it is not working within 6 to 8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment of depression is complete remission of current symptoms as well as prevention of future relapses
- The goal of treatment of chronic neuropathic pain is to reduce symptoms as much as possible, especially in combination with other treatments
- Treatment of depression most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Treatment of chronic neuropathic pain may reduce symptoms, but rarely eliminates them completely, and is not a cure since symptoms can recur after medicine is stopped
- Continue treatment of depression until all symptoms are gone (remission)
- Once symptoms of depression are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders and chronic pain may also need to be indefinite, but long-term treatment is not well studied in these conditions

If It Doesn't Work

- Many depressed patients have only a partial response where some symptoms are improved but others persist (especially insomnia,

fatigue, and problems concentrating)

- Other depressed patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- Lithium, buspirone, thyroid hormone (for depression)
- Gabapentin, tiagabine, other anticonvulsants, even opiates if done by experts while monitoring carefully in difficult cases (for chronic pain)

Tests

- Baseline ECG is recommended for patients over age 50
- * Monitoring of plasma drug levels is available

* Since tricyclic and tetracyclic antidepressants are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

* Monitor weight and BMI during treatment

* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antidepressant

- ECGs may be useful for selected patients (e.g., those with personal or family history of QTc prolongation; cardiac arrhythmia; recent myocardial infarction; uncompensated heart failure; or taking agents that prolong QTc interval such as pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin, etc.)
- Patients at risk for electrolyte disturbances (e.g., patients on diuretic therapy) should have baseline and periodic serum potassium and magnesium measurements

Side Effects

How Drug Causes Side Effects

- Anticholinergic activity may explain sedative effects, dry mouth, constipation, and blurred vision
- Sedative effects and weight gain may be due to antihistamine properties
- Blockade of alpha adrenergic 1 receptors may explain dizziness, sedation, and hypotension
- Cardiac arrhythmias and seizures, especially in overdose, may be caused by blockade of ion channels

Notable Side Effects

- Blurred vision, constipation, urinary retention, increased appetite, dry mouth, nausea, diarrhea, heartburn, unusual taste in mouth, weight gain
- Fatigue, weakness, dizziness, sedation, headache, anxiety, nervousness, restlessness
- Sexual dysfunction (impotence, change in libido)
- Sweating, rash, itching



Life-Threatening or Dangerous Side Effects

- Paralytic ileus, hyperthermia (TCAs + anticholinergic agents)
- Lowered seizure threshold and rare seizures